

Conservation of Energy (Recap)

SS ILP Physics (Intermediate), Week 4 Class 3



Outline

- Conservative Vs Non-conservative Force (Recap)
- Conservation of Mechanical Energy (Recap, with conservative force)
- General Law of Conservation of Energy (Recap, with non-conservative force)

Conservative Vs Non-conservative Force

- A conservative force allows conversion between kinetic and potential energy. Gravitational and the spring force are conservative.
- The work done between two points by any conservative force
 - I. is independent of the path between the two points.
 - II. is zero if the starting and ending points are the same.
 - III. is reversible.
 - IV. can be expressed in terms of a **potential energy function**. Note: only conservative force allows to define potential energy meaningfully due to property (I).
- A force (such as friction, air drag) that is not conservative is called a **non-conservative force**, or a *dissipative force*. Note: not all non-conservative forces are dissipative! e.g magnetic (Lorentz) force.

R.A. Serway and J.W. Jewett, Physics for Scientists and Engineers with Modern Physics. 9E, Cengage Learning, Boston, 2014.

Conservation of Mechanical Energy (Gravitational forces only, Recap)

From work-energy theorem,

$$W_{total} = \Delta K$$

We obtain

$$\Delta K = W_{total} = W_{grav} = -\Delta U_{grav}$$

$$K_2 - K_1 = -(U_{grav,2} - U_{grav,1})$$

$$K_1 + U_{grav,1} = K_2 + U_{grav,2}$$

$$\frac{1}{2}mv_1^2 + mgy_1 = \frac{1}{2}mv_2^2 + mgy_2$$

Total mechanical energy
at position y_1



Total mechanical energy
at position y_2

Conservation of mechanical energy:

When only the gravity does work, the total mechanical energy of the system is conserved:

$$E = K + U_{grav} = \text{constant}$$

H.D. Young and R. A. Freedman, Sears and Zemansky's
University Physics. 13E, Addison-Wesley, Boston, 2012.

Conservation of Mechanical Energy (Spring forces only, Recap)

From work-energy theorem,

$$W_{total} = \Delta K$$

We obtain

$$\Delta K = W_{total} = W_{el} = -\Delta U_{el}$$

$$K_2 - K_1 = -(U_{el,2} - U_{el,1})$$

$$K_1 + U_{el,1} = K_2 + U_{el,2}$$

$$\frac{1}{2}mv_1^2 + \frac{1}{2}kx_1^2 = \frac{1}{2}mv_2^2 + \frac{1}{2}kx_2^2$$

Total mechanical energy
at position x_1



Total mechanical energy
at position x_2

Conservation of mechanical energy:

When only the spring force does work, the total mechanical energy of the system is conserved:

$$E = K + U_{el} = \text{constant}$$

H.D. Young and R. A. Freedman, Sears and Zemansky's
University Physics. 13E, Addison-Wesley, Boston, 2012.

Conservation of Mechanical Energy

- Recall from previous lesson, the total mechanical energy is the summation of kinetic energy and potential energy:

$$E = K + U$$

- In the absence of non-conservative forces (we define this in next section), such as friction, total mechanical energy of an isolated system of objects is constant. It means that kinetic and potential energy simply converts into one another in the system.

$$E_i = E_f \quad \longleftrightarrow \quad K_i + U_i = K_f + U_f$$

- Or, if more than one type of potential energy involved

$$K_i + \sum U_i = K_f + \sum U_f$$

General Law of Conservation of Energy

- Friction and other non-conservative forces remove/add mechanical energy from/to the system. This is the amount by which the mechanical energy of the system changes, denoted by W_{other} . Thus,

$$K_i + U_i + W_{\text{other}} = K_f + U_f \quad \longleftrightarrow \quad E_i + W_{\text{other}} = E_f$$

- Example: In the case of kinetic friction and air resistance, since the force and displacement are always in opposite direction, the work done is always negative, and the final mechanical energy is always less than the initial mechanical energy (where d is distance travelled):

$$K_i + U_i - |\vec{F}_k|d = K_f + U_f \quad \longleftrightarrow \quad E_i - |\vec{F}_k|d = E_f$$

General Law of Conservation of Energy

- Non-conservative force cannot represent by potential energy, but we can describe the effects of these forces in terms of kinds of energy other than kinetic or potential energy.
- Example, when a car with locked brakes skids to a stop, both the tires and the road surface become hotter. The energy associated with this change in the state of the materials is called **Internal Energy** U_{int} .
- Experimentally, we observed that change in internal energy is:

$$\Delta U_{\text{int}} = -W_{\text{other}}$$

H.D. Young and R. A. Freedman, Sears and Zemansky's University Physics. 13E, Addison-Wesley, Boston, 2012.

General Law of Conservation of Energy

- We can write the law of conservation of energy as:

$$K_i + U_i - \Delta U_{\text{int}} = K_f + U_f \quad \longleftrightarrow \quad \Delta E + \Delta U_{\text{int}} = 0$$

Energy can never be created or destroyed. Energy may be transformed from one form to another, but the total energy of an isolated system is always constant!!!

Strategy for applying energy conservation

- Choose initial & final points.
- Select a zero-reference point for each form of P.E., and write the corresponding expression of P.E.
- If the net work done is only due to conservative force(s), apply

$$\boxed{E_i = E_f} \quad \longleftrightarrow \quad \boxed{K_i + \sum U_i = K_f + \sum U_f}$$

- If the net work done is (partly) due to non-conservative force(s) (like friction, air resistance), apply

$$\boxed{E_i + W_{\text{other}} = E_f \quad \text{or} \quad \Delta E + \Delta U_{\text{int}} = 0}$$

Concept Question

The spring in a pinball launcher has a spring constant of 405 N/m. How much work is required to compress the spring a distance of 3.00 cm?

- a) 0.091 J
- b) 0.182 J
- c) 0.273 J
- d) 0.364 J

Concept Question: Solution

The spring in a pinball launcher has a spring constant of 405 N/m. How much work is required to compress the spring a distance of 3.00 cm from equilibrium position?

- a) 0.091 J
- b) **0.182 J**
- c) 0.273 J
- d) 0.364 J

Set $U_1 = 0$, and $U_2 = \frac{1}{2}kx^2$ with $x = 3.00 \text{ cm}$.

By conservation of energy,

$$K_1 + U_1 + W_F = K_2 + U_2$$

$$0 + 0 + W_F = 0 + U_2$$

$$W_F = \frac{1}{2}kx^2 = 0.182 \text{ J}$$

Concept Question

When a force of 120.0 N is applied to a certain spring, it causes a stretch of 2.25 cm. What is the potential energy of this spring when it is compressed by 3.50 cm.

- a) 0.81 J
- b) 1.63 J
- c) 2.44 J
- d) 3.26 J

Concept Question: Solution

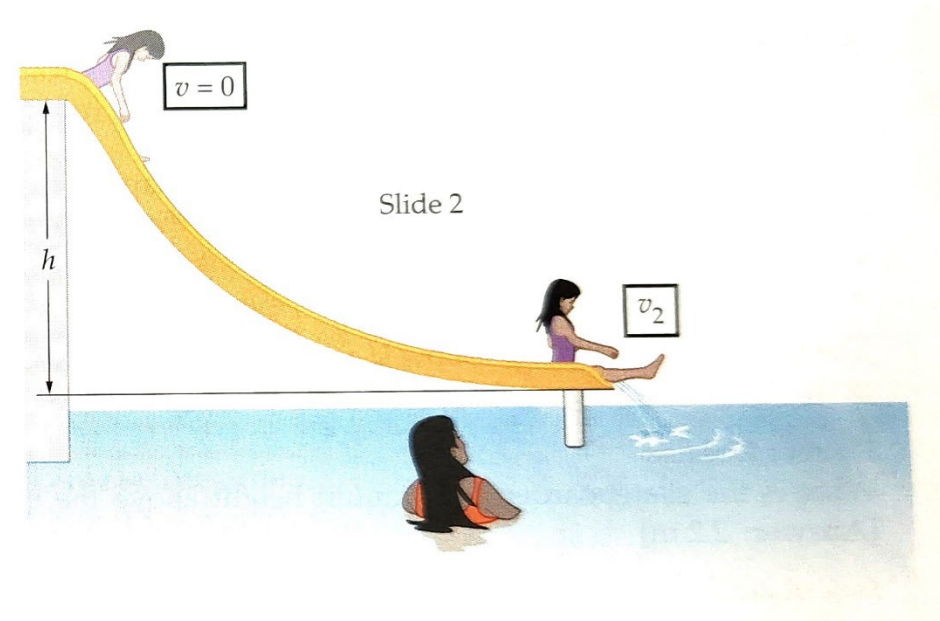
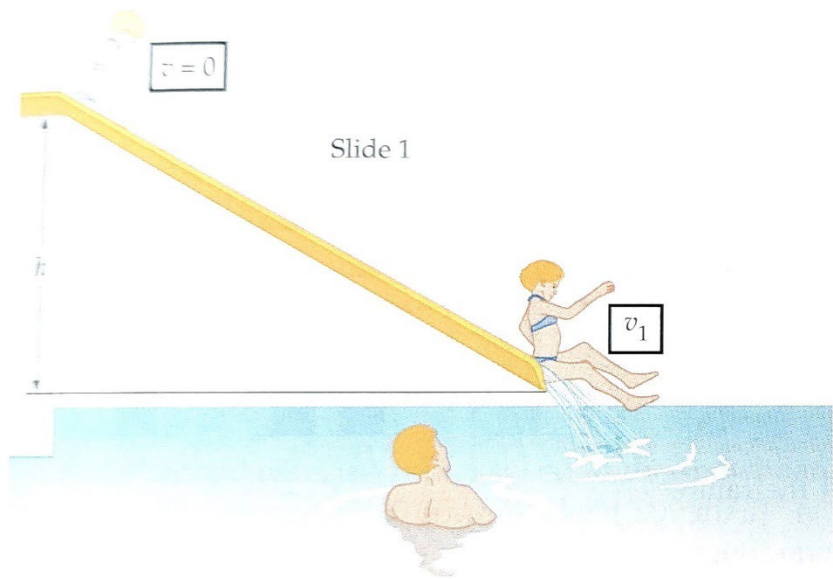
When a force of 120.0 N is applied to a certain spring, it causes a stretch of 2.25 cm. What is the potential energy of this spring when it is compressed by 3.50 cm.

- a) 0.81 J
- b) 1.63 J
- c) 2.44 J
- d) 3.26 J

$$F = kx \rightarrow k = \frac{F}{x} = 5330 \text{ N/m}$$
$$U = \frac{1}{2} kx^2 = 3.26 \text{ J}$$

Concept Question

Swimmers at a water park can enter a pool using one of two frictionless slides of equal height. Slide 1 approaches the water with a uniform slope, slide 2 dips rapidly at first, then levels out. Is the speed v_2 at the bottom of slide 2 (a) greater than, (b) less than, or (c) the same as the speed v_1 at the bottom of slide 1?



J. S. Walker, Physics, Conceptual Checkpoint 8-1, p219

Concept Question: Solution

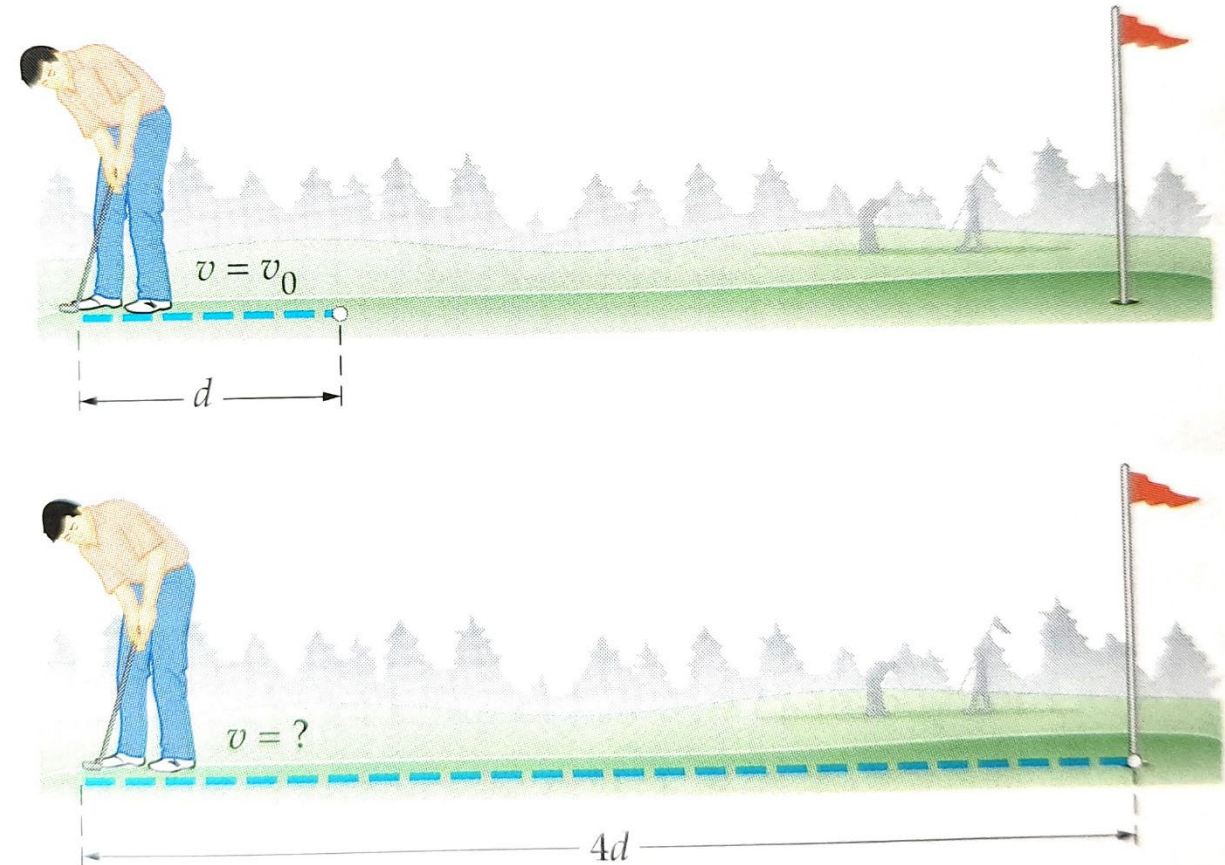
Swimmers at a water park can enter a pool using one of two frictionless slides of equal height. Slide 1 approaches the water with a uniform slope, slide 2 dips rapidly at first, then levels out. Is the speed v_2 at the bottom of slide 2 (a) greater than, (b) less than, or (c) **the same as** the speed v_1 at the bottom of slide 1?

By conservation of mechanical energy,

$$K_{1,i} + U_{1,i} = K_{1,f} + U_{1,f} \rightarrow 0 + mgh = \frac{1}{2}mv_1^2 + 0$$
$$K_{2,i} + U_{2,i} = K_{2,f} + U_{2,f} \rightarrow 0 + mgh = \frac{1}{2}mv_2^2 + 0$$

Concept Question

A golfer badly misjudges a putt, sending the ball only one-quarter of the distance to the pole. The original putt gave the ball an initial speed of v_0 . If the force of resistance due to the grass is constant, would an initial speed of (a) $2v_0$, (b) $3v_0$, or (c) $4v_0$ be needed to get the ball to the hole from its original position?



J. S. Walker, Physics, Conceptual Checkpoint 8-3, p225

Concept Question: Solution

A golfer badly misjudges a putt, sending the ball only one-quarter of the distance to the pole. The original putt gave the ball an initial speed of v_0 . If the force of resistance due to the grass is constant, would an initial speed of (a) $2v_0$, (b) $3v_0$, or (c) $4v_0$ be needed to get the ball to the hole from its original position?

By conservation of energy,

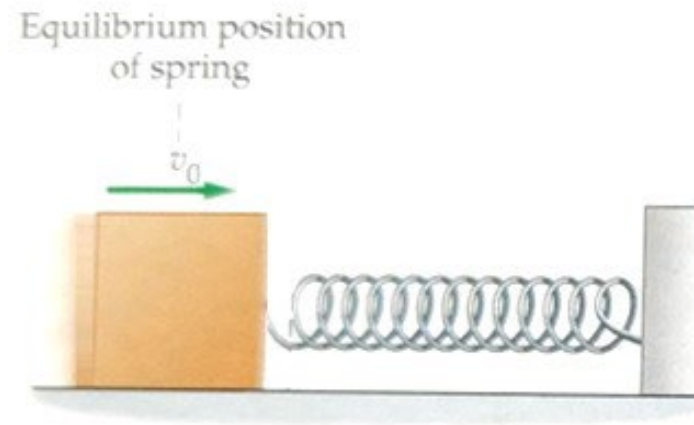
$$\begin{aligned}
 K_{1,i} + U_{1,i} + W_F &= K_{1,f} + U_{1,f} \\
 \frac{1}{2}mv_0^2 + 0 + W_{1,F} &= 0 + 0 \\
 \frac{1}{2}mv_0^2 - \frac{Fd}{4} &= 0
 \end{aligned}$$

By conservation of energy,

$$\begin{aligned}
 K_{2,i} + U_{2,i} + W_F &= K_{2,f} + U_{2,f} \\
 \frac{1}{2}mv^2 + 0 + W_{2,F} &= 0 + 0 \\
 \frac{1}{2}mv^2 - Fd &= 0
 \end{aligned}$$

Case Problem 1

A block has a mass of 1.5 kg and slides to right on a frictionless surface with a speed of $v_0 = 2.2$ m/s from equilibrium position. Find the compression of the spring, which spring constant is 475 N/m, when the block momentarily comes to rest.



J. S. Walker, Physics, Active Example 7-1, p194

Case Problem 1: Solution

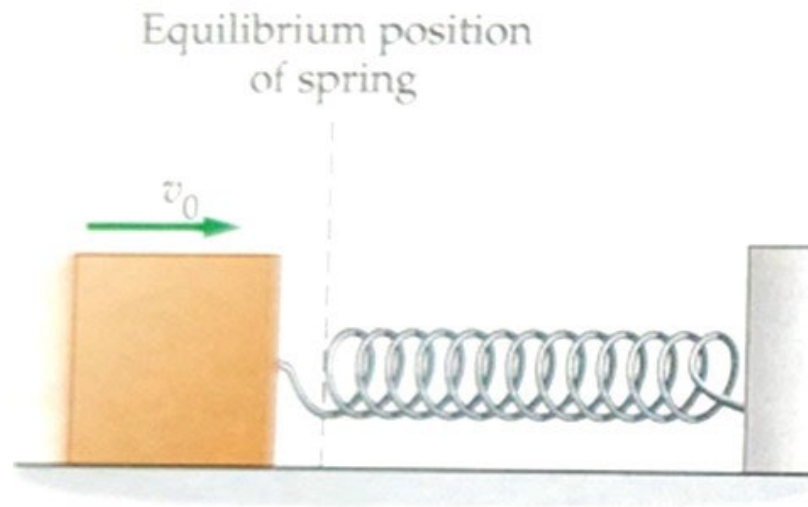
A block has a mass of 1.5 kg and slides to right on a frictionless surface with a speed of $v_0 = 2.2$ m/s from equilibrium position. Find the compression of the spring, which spring constant is 475 N/m, when the block momentarily comes to rest.

From conservation of mechanical energy,

$$\begin{aligned} K_i + U_i &= K_f + U_f \\ \frac{1}{2}mv_0^2 + 0 &= 0 + \frac{1}{2}kx_f^2 \\ x_f &= 0.12 \text{ m} \end{aligned}$$

Case Problem 2

A block has a mass of 1.5 kg and slides to right on a frictionless surface with a speed of $v_0 = 2.2$ m/s from position A which is at a distance of 0.1 m from equilibrium position, as shown. Find the compression of the spring, which spring constant is 475 N/m, when the block momentarily comes to rest.



J. S. Walker, Physics, Active Example 7-1, p194

Case Problem 2: Solution

A block has a mass of 1.5 kg and slides to right on a frictionless surface with a speed of $v_0 = 2.2$ m/s from position A which is at a distance of 0.1 m from equilibrium position, as shown. Find the compression of the spring, which spring constant is 475 N/m, when the block momentarily comes to rest.

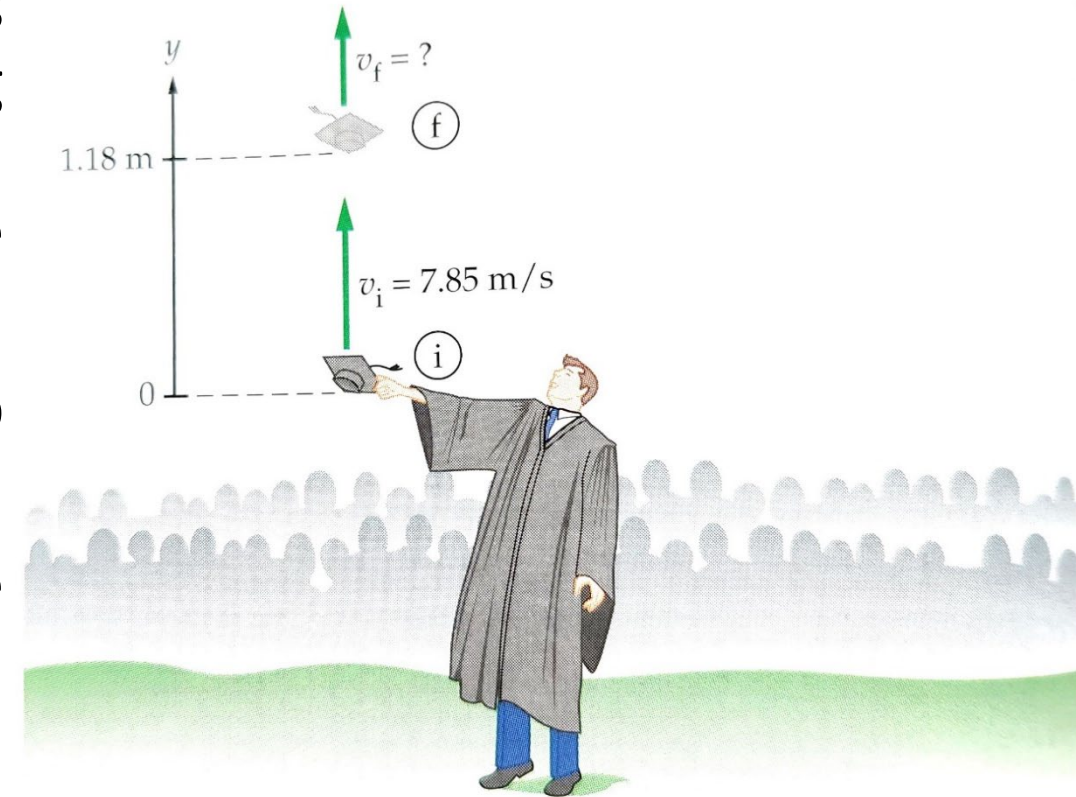
From conservation of mechanical energy,

$$\begin{aligned} K_i + U_i &= K_f + U_f \\ \frac{1}{2}mv_0^2 + \frac{1}{2}kx_0^2 &= 0 + \frac{1}{2}kx_f^2 \\ x_f &= 0.16 \text{ m} \end{aligned}$$

Case Problem 3

At the end of a graduation ceremony, graduates fling their caps into the air. Suppose a 0.12-kg cap is thrown straight upward with an initial speed of 7.85 m/s and air resistance can be neglected. Take $g = 9.81 \text{ m/s}^2$.

- Use kinematics to find the speed of the cap when it is 1.18 m above the release point.
- Show that the mechanical energy at the release point is the same as the mechanical energy 1.18 m above the release point.



Case Problem 3 : Solution

At the end of a graduation ceremony, graduates fling their caps into the air. Suppose a 0.12-kg cap is thrown straight upward with an initial speed of 7.85 m/s and air resistance can be neglected. Take $g = 9.81 \text{ m/s}^2$.

- a) Use kinematics to find the speed of the cap when it is 1.18 m above the release point.

$$v_y^2 - v_{y,0}^2 = -2g (y - y_0)$$

$$v_y = \sqrt{v_{y,0}^2 - 2g (y - y_0)} = 6.20 \text{ m/s}$$

Case Problem 3: Solution

At the end of a graduation ceremony, graduates fling their caps into the air. Suppose a 0.12-kg cap is thrown straight upward with an initial speed of 7.85 m/s and air resistance can be neglected. Take $g = 9.81 \text{ m/s}^2$.

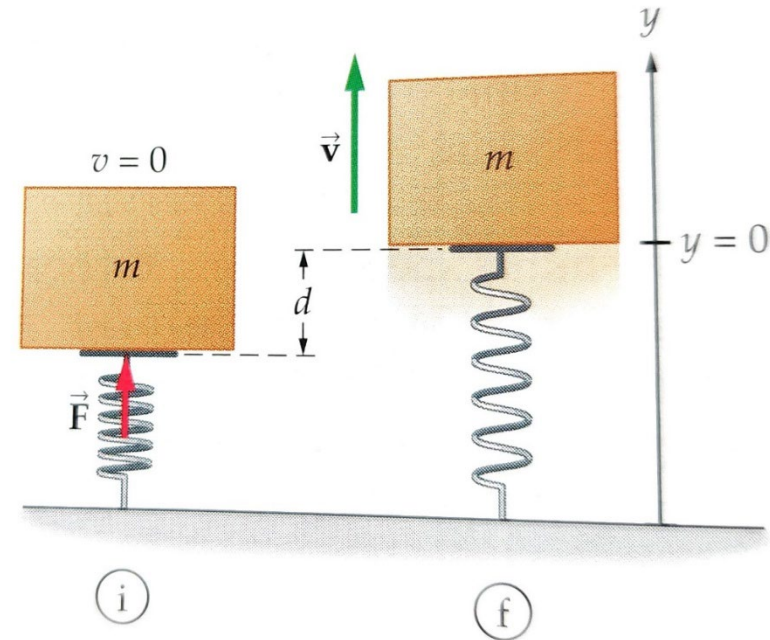
b) Show that the mechanical energy at the release point is the same as the mechanical energy 1.18 m above the release point.

$$E_i = K_i + U_i = \frac{1}{2}mv_i^2 + 0 = \frac{1}{2}(0.12 \text{ kg})(7.85 \text{ m/s})^2 = 3.70 \text{ J}$$

$$\begin{aligned} E_f &= K_f + U_f = \frac{1}{2}mv_f^2 + mgh \\ &= \frac{1}{2}(0.12 \text{ kg})(6.20 \text{ m/s})^2 + (0.12 \text{ kg})(9.81 \text{ m/s}^2)(1.18 \text{ m}) = 3.70 \text{ J} \end{aligned}$$

Case Problem 4

Given a spring (constant of spring is 955 N/m) and a block (1.7 kg) that are oriented vertically, as shown. Initially, the spring is compressed 4.6 cm and the block is at rest. When the block is released, it accelerates upward. Find the speed of the block when the spring has returned to its equilibrium position. Take $g = 9.81 \text{ m/s}^2$.



J. S. Walker, Physics, Active Example 8-1, p222

Case Problem 4: Solution

Given a spring (constant of spring is 955 N/m) and a block (1.7 kg) that are oriented vertically, as shown. Initially, the spring is compressed 4.6 cm and the block is at rest. When the block is released, it accelerates upward. Find the speed of the block when the spring has returned to its equilibrium position. Take $g = 9.81 \text{ m/s}^2$.

From conservation of mechanical energy,

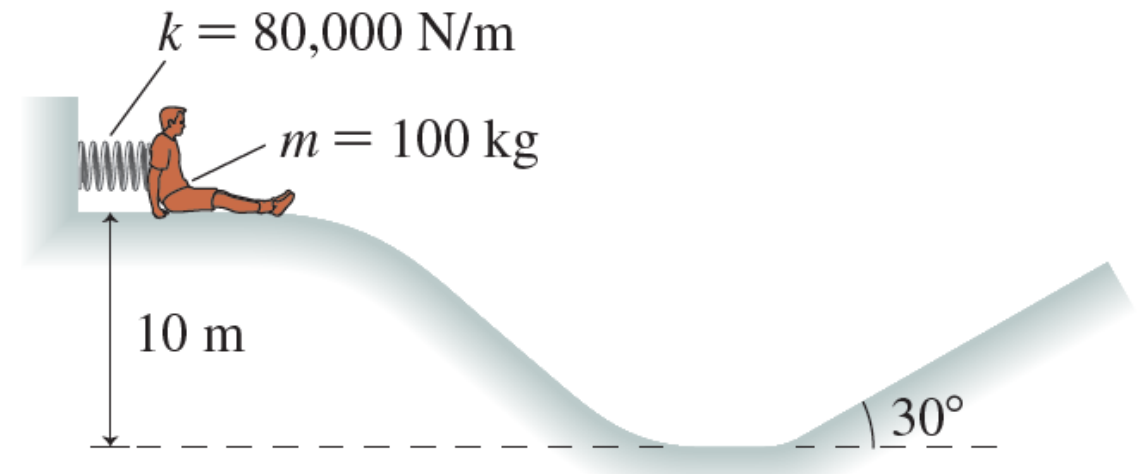
$$\begin{aligned} K_i + U_i &= K_f + U_f \\ 0 + \frac{1}{2}ky_i^2 - mgd &= \frac{1}{2}mv_f^2 + 0 \\ v_f &= \sqrt{\frac{ky_i^2 - 2mgd}{m}} = 0.535 \text{ m/s} \end{aligned}$$

Off-class practice question(s)

Case Problem 5:

The spring (with spring constant $k = 80,000 \text{ N m}^{-1}$) shown below is compressed 50 cm and used to launch a 100 kg physics student in ILP class. The track is frictionless until it starts up the incline. The student's coefficient of kinetic friction on the 30° incline is $\mu_k = 0.15$.

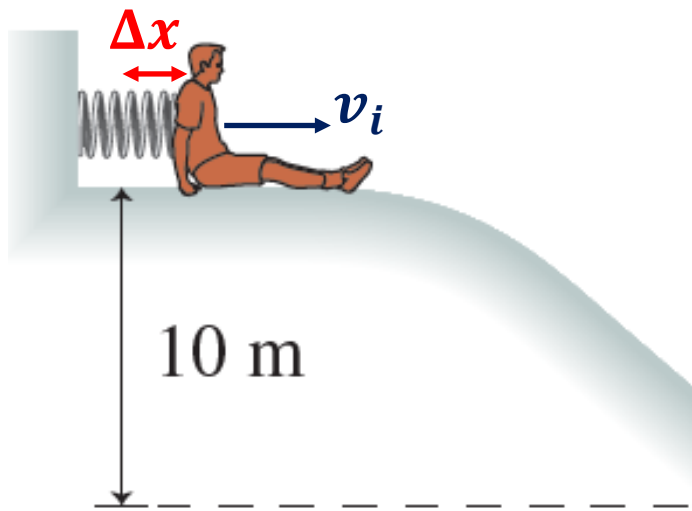
- What is the student's speed just after losing contact with the spring?
- How far up the incline does the student go?



R.D. Knight, Physics for Scientists and Engineers: A Strategic Approach. 4E, Pearson, Boston, 2016.

Case Problem 5: Solution

a). Consider phase 1 where spring is going from compressed to uncompressed state, the elastic potential energy all converted to the kinetic energy of the person. Assume spring is ideal and its mass is negligible compared to the person. We have,



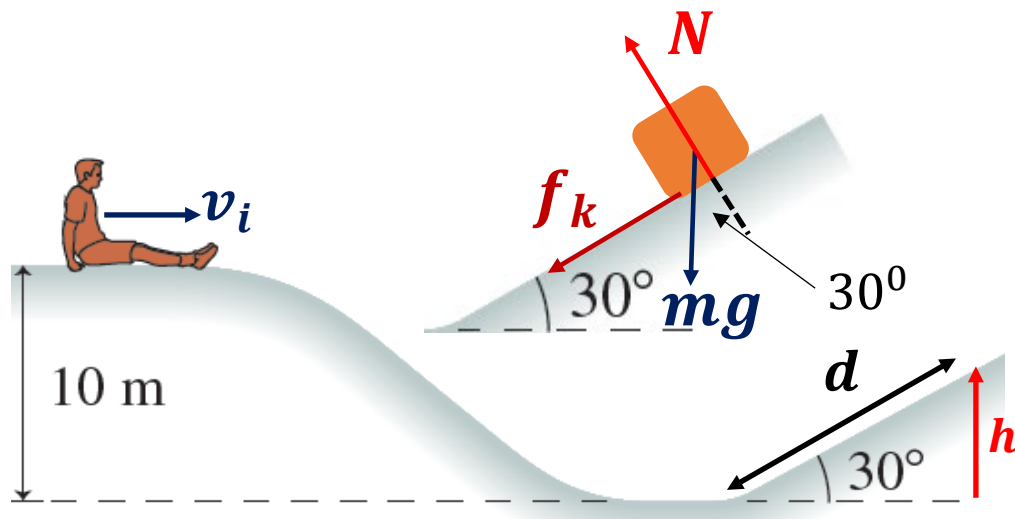
$$\frac{1}{2}k(\Delta x)^2 = \frac{1}{2}mv_i^2 \Rightarrow v_i = \sqrt{\frac{k}{m}} \cdot \Delta x$$

$$v_i = \sqrt{\frac{8 \times 10^4 \text{ Nm}^{-1}}{100 \text{ kg}}} \cdot (0.50 \text{ m}) = 14 \text{ ms}^{-1}$$

R.D. Knight, Physics for Scientists and Engineers:
A Strategic Approach. 4E, Pearson, Boston, 2016.

Case Problem 5: Solution

b). Phase 2: The person lost contact with the spring and move long the track. Let f be the final point along the incline plane where the person stop. Apply the general law of conservation of energy, $K_i + U_i + W_{\text{other}} = K_f + U_f$ to the person. From the question $K_f = 0$ (finally at stop), the only non-conservative force is the friction between the person with the inclined plane, so we have $W_{\text{other}} = -|\vec{f}_k|d = -\mu_k mg \cos 30^\circ d$.



From COE and using trigonometric, $h = d \sin 30^\circ$,

$$\frac{1}{2}mv_i^2 + mgh_i - \mu_k mg \cos 30^\circ d = 0 + mgh$$

$$\Rightarrow d = \frac{v_i^2/2 + gh_i}{\mu_k g \cos 30^\circ + g \sin 30^\circ} = 32 \text{ m}$$

R.D. Knight, Physics for Scientists and Engineers: A Strategic Approach. 4E, Pearson, Boston, 2016.